

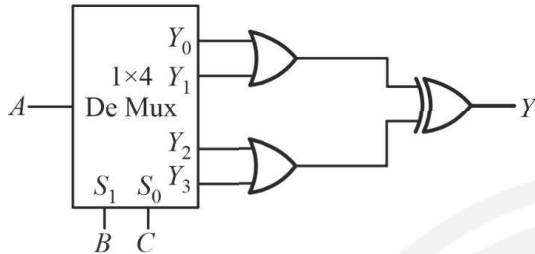
DPP-01

CS & IT

Digital Logic

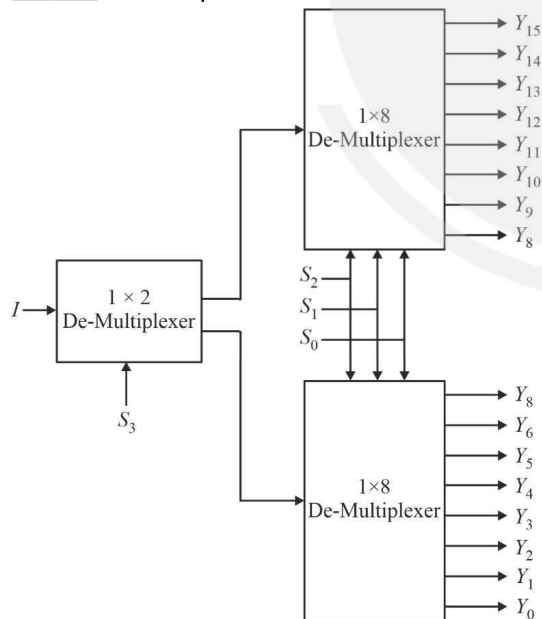
Combinational Circuit

Q1 For what values of A, B, C the output (Y) will be 0



- (A) $A = 1, B = 0, C = 0$
- (B) $A = 0, B = 1, C = 1$
- (C) $A = 1, B = 1, C = 0$
- (D) $A = 1, B = 1, C = 1$

Q2 The figure shown below is a block diagram of _____ demultiplexer?



- (A) 1 to 4
- (B) 1 to 8
- (C) 1 to 16

(D) None of the above

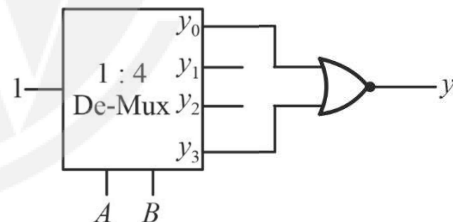
Q3 How many inputs will a decimal to BCD encoder have? _____

Q4 Which one of the following de multiplexer requires only five select lines?

- (A) 1×2 de Mux
- (B) 1×4 De Mux
- (C) 1×8 De Mux
- (D) 1×32 De Mux

Q5 What is the minimum number of 1×4 De Mux required to implement 1×2^{10} De Mux. _____

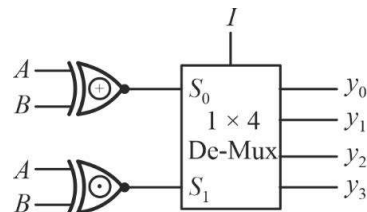
Q6 Consider a circuit as shown below:



Output y is

- (A) $A + B$
- (B) $\overline{A \cdot B}$
- (C) $A \oplus B$
- (D) $A \odot B$

Q7 Consider a combinational circuit as shown below.



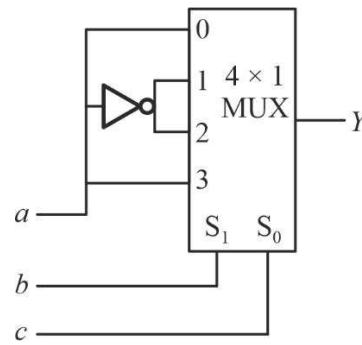
For any sequence A, B which of the output pins (y_0 to y_3) can be active

- (A) y_0 and y_3 only
- (B) y_1 and y_2 only
- (C) y_1 only
- (D) all pins can be active

Q8 What are basic gates required to implement a full adder

- (A) 1 EX-OR gate, 1 AND gate
- (B) 2 EX-OR gate, 1 OR gate
- (C) 2 EX-OR gate, 2 AND gate, 1 OR gate
- (D) 1 EX-OR gate, 2 AND gate, 2 OR gate

Q9 The following multiplexer circuit is equivalent to _____.



- (A) Implementation of sum equation of full adder
- (B) Implementation of carry equation of full adder
- (C) Implementation of borrow equation of full subtractor
- (D) All the above

Q10 Eight 1-bit full adders are cascaded. Each 1-bit full adder generates carry out bit in 10 ns and sum bit in 30 ns. The number of addition performed per second, will be _____ $\times 10^7$.



Answer Key

Q1 (B)

Q2 (C)

Q3 10~10

Q4 (D)

Q5 341~341

Q6 (C)

Q7 (B)

Q8 (C)

Q9 (A)

Q10 1~1



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